Q1

A ship's waterplane is 18metres long. The half-ordinates at equal distances from forward are as follows:

0, 1.2, 1.5, 1.8, 1.8, 1.5 and 1.2 metres, respectively

Find the second moment of the waterplane area about the centreline.

10marks

Q2

When the vertical centre of gravity G rises, there will normally be a loss in the ship's stability. G may even rise above the transverse metacentre M to make the ship unstable. The master and mate onboard ship must be aware of changes in a ship that would cause such a rise in G.
Gives Twenty(20) reasons for such a rise.
10marks

Q3

A ship arrives in port with displacement 6000 tonnes and KG 6 m. She then discharges and loads the following quantities:

Discharge
1250 tonnes of cargo KG 4.5 metres
675 tonnes of cargo KG 3.5 metres
420 tonnes of cargo KG 9.0 metres

Load
980 tonnes of carrgo KG 4.25 metres
550 tonnes of cargo KG 6.00 metres

700 tonnes of bunkers KG 1.0 metre
70 tonnes of FW KG 12.0 metres

During the stay in port 30 tonnes of oil (KG 1 m) are consumed. If the final KM is 6.8 m, find the GM on departure.

Handing in via google classroom
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